



CONFIDENTIAL TECHNICAL REPORT

Maintenance & Health Check (MHC) Report

System Health, Calibration & Optical Performance Assessment

MACHINE & CUSTOMER IDENTITY

MHC EVALUATION & ASSESSMENT

CUSTOMER ACCOUNT
Acme PCB Corp

PLANT / FACILITY
Plant 2 - Penang

PRODUCTION LINE
—

ZONE
—

MACHINE MODEL
ESI 5330 Flex

MACHINE NUMBER / SOURCE
WLVIA#3

SERIAL NUMBER
SN-98765

BASELINE DATE
—

LAST MHC DATE
2026-09-02

SERVICE ENGINEER
Sahafiz
Field Service Engineer

INSPECTION DATE
2026-09-02

RELEASE STATUS
⌚ PENDING REVIEW

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● MAINTENANCE & HEALTH CHECK REPORT STRUCTURE

This formal technical report presents authoritative inspection, telemetry, and calibration records collected for this equipment. Subsystems are documented with baseline comparisons and recorded evidence where applicable.

03 EXECUTIVE SUMMARY

MHC EVALUATION

MHC RESULT

ACTION REQUIRED – 11%

ACTION_REQUIRED

Routine Maintenance & Health Check (MHC) completed for Flex Drill WLVI#3 (SN-98765) at Acme PCB Corp - Plant 2 - Penang.

MHC INSPECTION RESULTS

SUBSYSTEM / AUDIT ITEM	SPECIFICATION	VERDICT
Laser Power (Laser Head 1 & 2)	15.0 W ± 10% (13.5–16.5 W)	WARNING
Beam Profile / Mode	–	WARNING
Stage Calibration	–	– NOT COLLECTED
AGC / Scanner Calibration	–	– NOT COLLECTED
Temperature & Cooling	–	– NOT COLLECTED

04 LASER LIFECYCLE & USAGE TELEMETRY

1 HEADS ANALYZED

Laser Head 1

PASS



LIFECYCLE PROGNOSIS & SERVICE RECOMMENDATIONS

PROGNOSTIC ADVISORY

Laser Head 1: Mid-to-late life phase (28.7% remaining). Tube capacity is stable. Monitor optical power decay during regular service.

05 LASER POWER & BASELINE

COMPARISON

Authoritative Optical Power Measurement & Calibration
Variance Analysis

SPEC: 15.0W ± 10% (13.50-16.50 W)

NOT COLLECTED

● LEFT: Historical Baseline (Previous)

VS ● RIGHT: Present Measurement (Current)

→ ● VARIATION: Δ = Current - Previous

Laser Head 1

SPEC: 15.00 W ± 10% (13.50-16.50 W)

WARNING

PREVIOUS (BASELINE)

Baseline

—

Historical Baseline

VS
COMPARE

CURRENT (PRESENT)

2026-09-02

Not Recorded

Target: 15.00 W

VARIATION = CURRENT - PREVIOUS

Calibration Shift:

No previous baseline

Laser Head 2

SPEC: 15.00 W ± 10% (13.50-16.50 W)

WARNING

PREVIOUS (BASELINE)

Baseline

—

Historical Baseline

VS
COMPARE

CURRENT (PRESENT)

2026-09-02

Not Recorded

Target: 15.00 W

VARIATION = CURRENT - PREVIOUS

Calibration Shift:

No previous baseline

06 OPTICAL BEAM PROFILE & SPOT QUALITY

Spatial Beam Distribution & Multistage Aperture Quality Telemetry

PASS

LASER HEAD 1

Laser Head 1

Baseline: 3.500 mm | Δ Variation: +0.000 mm (0.0%)

PASS

1. LASER SOURCE (6A)

PASS

CURRENT MEASURED SPOT SIZE

Ø 3.500 mm

APPLICABLE SPECIFICATION:
3.5 mm ±10% (3.15-3.85 mm)

2. FLAT TOP (6B)

PASS

CURRENT MEASURED SPOT SIZE

Ø 4.150 mm

APPLICABLE SPECIFICATION:
4.2 mm ±5% (3.99-4.41 mm)

APERTURE MASK PROFILE EVIDENCE (0.9 MM – 2.2 MM)

6 / 6 Within Target Spec

0.9mm

PASS

Ø 0.950 mm
≥0.9mm

1.1mm

PASS

Ø 1.150 mm
≥1.1mm

1.3mm

PASS

Ø 1.350 mm
≥1.3mm

1.8mm

PASS

Ø 1.850 mm
≥1.8mm

2.0mm

PASS

Ø 2.050 mm
≥2.0mm

2.2mm

PASS

Ø 2.250 mm
≥2.2mm

LASER HEAD 2

Laser Head 2

Baseline: 3.500 mm | Δ Variation: +0.000 mm (0.0%)

PASS

1. LASER SOURCE (7A)

PASS

CURRENT MEASURED SPOT SIZE

Ø 3.500 mm

APPLICABLE SPECIFICATION:
3.5 mm ±10% (3.15-3.85 mm)

2. FLAT TOP (7B)

PASS

CURRENT MEASURED SPOT SIZE

Ø 4.150 mm

APPLICABLE SPECIFICATION:
4.2 mm ±5% (3.99-4.41 mm)

APERTURE MASK PROFILE EVIDENCE (0.9 MM – 2.2 MM)

6 / 6 Within Target Spec

0.9mm

PASS

Ø 0.950 mm
≥0.9mm

1.1mm

PASS

Ø 1.150 mm
≥1.1mm

1.3mm

PASS

Ø 1.350 mm
≥1.3mm

1.8mm

PASS

Ø 1.850 mm
≥1.8mm

2.0mm

PASS

Ø 2.050 mm
≥2.0mm

2.2mm

PASS

Ø 2.250 mm
≥2.2mm

07 FOCUS OPTIMIZATION

Machining Focus Calibration & Focal Sequence Verification

FOCUS ADJUSTMENT DATE
2026-09-02

ADJUSTMENT REASON
Laser source replacement

BASELINE
-0.300 mm

EVALUATION
Laser source optical alignment verified via dummy wafer drill matrix. Focus set to 0.00 position.

LASER HEAD 1

BASELINE: -0.300 mm

+3

Laser Head 1 Wafer Drill

+0.300 mm
Focus Position

+2

Laser Head 1 Wafer Drill

+0.200 mm
Focus Position

+1

Laser Head 1 Wafer Drill

+0.100 mm
Focus Position

0

Laser Head 1 Wafer Drill

0.000 mm
Focus Position

-1

Laser Head 1 Wafer Drill

-0.100 mm
Focus Position

-2

Laser Head 1 Wafer Drill

-0.200 mm
Focus Position

-3

Laser Head 1 Wafer Drill

-0.300 mm
Focus Position

LASER HEAD 2

BASELINE: -0.300 mm

+3

Laser Head 2 Wafer Drill

+0.300 mm
Focus Position

+2

Laser Head 2 Wafer Drill

+0.200 mm
Focus Position

+1

Laser Head 2 Wafer Drill

+0.100 mm
Focus Position

0

Laser Head 2 Wafer Drill

0.000 mm
Focus Position

-1

Laser Head 2 Wafer Drill

-0.100 mm
Focus Position

-2

Laser Head 2 Wafer Drill

-0.200 mm
Focus Position

-3

Laser Head 2 Wafer Drill

-0.300 mm
Focus Position

Note: Top via impact: Focus adjustment primarily affects top diameter (~90%), significantly influencing top via.

08 POWER OFFSET & CALIBRATION

Power Offset Range: -20% to +20%

LASER HEAD 1 (LH1)

POWER OFFSET: —

PHASE 1

Recipe: —

Adjusted: —

PHASE 2

Recipe: —

Adjusted: —

OFFSET COMPARISON

ADJUSTMENT REASON

—

Prev: — Curr: —

LASER HEAD 2 (LH2)

POWER OFFSET: —

PHASE 1

Recipe: —

Adjusted: —

PHASE 2

Recipe: —

Adjusted: —

OFFSET COMPARISON

ADJUSTMENT REASON

—

Prev: — Curr: —

Note: Bottom via impact: Power offset primarily influences bottom via diameter (~90%), making it a key factor in bottom via diameter control.

09 STAGE CALIBRATION (X/Y DEVIATION)

Sub-Micron Motion Stage Positional Accuracy & Cross-Axis Verification

TOLERANCE: $\pm 2.0 \mu\text{m}$

NOT COLLECTED

STAGE IDENTIFIER	X DEVIATION RANGE [MEASURED]	Y DEVIATION RANGE [MEASURED]	MAX ABS DEV [DERIVED]	VERDICT
Stage 1 (Left)	—	—	—	UNANSWERED
Stage 2 (Right)	—	—	—	UNANSWERED
Calibration Record: Stage positional calibration completed across the full travel range, with X/Y deviation verified against the defined tolerance.				

10 AGC / SCANNER CALIBRATION

Galvo Scanner & Automatic Grid Calibration Positional Verification

TOLERANCE: $\pm 3.0 \mu\text{m}$

NOT COLLECTED

AGC IDENTIFIER	X DEVIATION RANGE [MEASURED]	Y DEVIATION RANGE [MEASURED]	VERDICT
AGC 1	—	—	UNANSWERED
AGC 2	—	—	UNANSWERED
Calibration Record: AGC/scanner positional calibration completed across the full travel range, with X/Y deviation verified against the defined tolerance.			

11 TEMPERATURE & THERMAL TELEMETRY

— NOT COLLECTED

Optics Markbox 6-Channel Air-Cooling & Thermal Telemetry Monitoring

OPTICS MARKBOX AIR-COOLING SUBSYSTEM OVERVIEW

TARGET SPEC: —

MARKBOX 1NO DATA

Not Linked

Optics Markbox 1 (CH1 + CH4)

MARKBOX 2NO DATA

Not Linked

Optics Markbox 2 (CH2 + CH5)

MARKBOX 3NO DATA

Not Linked

Optics Markbox 3 (CH3 + CH6)

NO CONTINUOUS MULTI-CHANNEL THERMAL LOG LINKED TO THIS SESSION

To populate detailed 6-channel optics markbox sensor matrices and profile curves,
link a temperature log record to this machine.

12 PRODUCT PROCESS & VIA QUALITY

— NOT COLLECTED

Product Recipe Parameters & Via Measurements

PRODUCT
Not Recorded

RECIPE / PROGRAM
Not Recorded

LOT / PANEL IDENTIFIER
Not Recorded

PROCESS RECIPE PHASE PARAMETERS

PROCESS PHASE	POWER (W)	FREQUENCY (kHz)	SHOT COUNT	MASK (mm)	DEFOCUS (mm)
Phase 1	-	-	-	-	-
Phase 2	-	-	-	-	-

VIA DRILLING MEASUREMENTS & SPECIFICATION

LASER HEAD	TOP DIA.	BOT DIA.	TAPER	VERDICT
Acceptance Spec	-	-	-	Engineering Spec
Laser Head 1 (LH1)	-	-	-	NOT COLLECTED
Laser Head 2 (LH2)	-	-	-	NOT COLLECTED

VIA CROSS-SECTION EVIDENCE

No Image

Laser Head 1 (LH1)

No Image

Laser Head 2 (LH2)

Engineer Remarks: —

13 FINDINGS & OBSERVATIONS

TOTAL RECORDED: 0

— NOT COLLECTED

—

ALL SUBSYSTEMS NOMINAL

14 SPARE PARTS & RECOMMENDATIONS

— NOT COLLECTED

ENGINEERING RECOMMENDATIONS & FUTURE ACTION PLAN

—

CONSUMED PARTS (SERVICE EXECUTION)

—

RECOMMENDED SPARE PARTS (PROCUREMENT / STOCK)

—

15 BUYOFF & OFFICIAL APPROVALS

PRODUCTION RELEASE: PENDING

🕒 PENDING REVIEW

NEXT SCHEDULED MHC

Target Due Date: 2026-12-01

Cycle: 90-Day Standard Interval

FIELD SERVICE ENGINEER

VERIFIED

Sahafiz

Field Service Engineer

Date: 2026-09-02

[COMPLETED & SIGNED BY ENGINEER]

CUSTOMER ACCEPTANCE REPRESENTATIVE

PENDING

Pending Customer Sign-off

Acme PCB Corp

Date: —

[PENDING CUSTOMER REVIEW & SIGN-OFF]